



Improving the probabilistic drought prediction with soil moisture information under the ensemble streamflow prediction framework

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Abstract

Reliable drought prediction should be preceded to prevent damage from potential droughts. In this context, this study developed a hydrological drought prediction method, namely ensemble drought prediction (EDP) to reflect drought-related information under the ensemble streamflow prediction framework. After generating an ensemble of standardized runoff index by converting the ensemble of generated streamflow, the results were adopted as the prior distribution. Then, precipitation forecast and soil moisture were used to update the prior EDP. The EDP + A model included the precipitation forecast with the PDF-ratio method, and the observed soil moisture index was reflected in the former EDP and EDP + A via Bayes' theorem, resulting in the EDP + S and EDP + AS models. Eight basins in Korea with more than 30 years of observation data were applied with the proposed methodology. As a result, the overall performance of the four EDP models yielded improved results than the climatological prediction. Moreover, reflecting soil moisture yielded improved evaluation metrics during short-term drought predictions, and in basins with larger drainage areas. Finally, the methodology presented in this study was more effective during periods with less intertemporal variabilities.

Keywords Probabilistic drought prediction · Standardized runoff index · Soil moisture · Ensemble streamflow prediction · Bayesian update

1 Introduction

It is estimated that about two-thirds of the world's population has experienced severe water scarcity (Mekonnen and Hoekstra 2016), and the dry will get even more drier as global warming has accelerated the hydrological cycle resulting in more extremes (Seager et al. 2010; Trenberth et al. 2014). Although droughts are complex catastrophes that are difficult to classify into a single category, review studies on droughts often classify droughts into the following four categories: meteorological, agricultural, hydrological, and socio-economic (Ndayiragije and Li 2022). Whereas

meteorological and hydrological droughts are associated with physical phenomena (i.e., rainfall, surface water, and groundwater), the other two classifications focus more on other aspects of droughts, such as soil moisture and societal perspectives.

According to Korean National Drought Information-analysis Center, droughts have also been occurring locally in Korea for almost every year since 2006, and a record-breaking multi-year drought from 2014 through 2019 due to the lack of precipitation on the western coast of the Korean Peninsula (Kim et al. 2019, 2021) have heavily affected the lives of the public. In response, multiple efforts for probabilistic prediction in drought conditions have been conducted to quantify the uncertainty and derive results that can help better decision-making. Novel probabilistic drought forecasting models, such as the Bayesian approach (Madadgar and Moradkhani 2013, 2014) were often used to forecast future droughts preserving spatial-temporal variations. More recently, utilizing diverse types of data such as remotely sensed data and satellite data for drought monitoring and impact assessments (Gavahi et al. 2020; Li et al. 2019; Park

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et al. 2018) were proven to enhance drought predictions in various time scales.

A case in point is the ensemble streamflow prediction (ESP) framework, which is one of the most widely utilized and studied tools in the hydrology field over decades (Day 1985). ESP generates the ensemble of streamflow forecasts with the previous historical meteorological observation data given identical initial conditions. The purpose of the ESP is not just to represent an error bar around the predicted values, but to bundle up possible scenarios in the future to estimate probabilities of events. Branković et al. (1990) proved that the probabilistic prediction via ESP has better accuracy than a single-valued prediction. However, since the traditional ESP are not able to reflect the actual hydrometeorological conditions at the time of interest, multiple approaches have applied weather predictions generated by dynamic models of climate and ocean system to the ESP (Saha et al. 2014; Yuan et al. 2014). Harrigan et al. (2018) found that the skill of the ESP decreases exponentially with increasing lead times even using a variety of climate ensembles because the uncertainties in initial hydrologic conditions (e.g., soil moisture) still exist. Recent novel approaches to the application of ESP also include snow data assimilation to enhance predictions (DeChant and Moradkhani 2011) and utilizing machine learning algorithms to improve prediction reliability (Tyralis et al. 2021; Liu et al. 2022).

As the traditional ESP framework was adopted to predict streamflow, the conversion of streamflow prediction ensembles into an appropriate drought measure such as a drought index, should be preceded to utilize ESP during probabilistic drought forecasts. Consequentially, few studies endeavored to convert the precipitation ensemble into the standardized precipitation index (SPI), which is one the most widely used drought indices, to predict meteorological drought (Yoon et al. 2012; Mo and Lyon 2015). In addition, Son and Bae (2015) tested a model that produces standardized runoff index (SRI) ensembles by converting ESP simulation into the SRI for hydrological drought prediction and later improved the skill of the model using climate dynamic models and Bayesian methods (Bae et al. 2017). This Bayesian method combines empirical knowledge with numerical analysis results through Bayesian inferences for more accurate predictions. This approach has also been used to improve the simulation performance of hydrologic conditions (Luo et al. 2007). However, estimating hydrological droughts with only streamflow forecast ensembles may yield inaccurate results since many other factors substantially affect the hydrological process, as highlighted in previous literature.

As soil moisture is often considered one of the most important factors during the hydrological processes, multiple studies have been carried out continuously to make

accurate estimates of soil moisture and to analyze its effects on agricultural droughts (Wood and Lettenmaier 2008; Li et al. 2009; Shukla and Lettenmaier 2011; Koster et al. 2014; Yuan et al. 2016). Moreover, it was also found that precipitation forcing has a significant impact on the uncertainties of predictions of hydrological drought and soil moisture (Mo et al. 2012). Although it has been verified that the relationships between hydrological processes and the conditions of climate and soil moisture are significantly important in drought assessment in multiple previous studies, few efforts exist that reflect the information from the drought-related factors (soil moisture and climatic information) during the hydrological drought predictions.

Thus, this study proposes an ensemble drought prediction (EDP) framework, which converts the streamflow simulation ensembles from the conventional ESP framework into a representative drought index for hydrological drought outlook. Then, the probabilistic precipitation forecasts and the information regarding soil moisture were used to update the previous forecasts via Bayes' Theorem to improve prediction accuracy. The proposed methodology was then applied to eight major watersheds in South Korea, and the results were evaluated in terms of prediction accuracy with both deterministic and probabilistic metrics.

2 Methodology

2.1 Ensemble streamflow prediction

With the assumption that the historical records may potentially represent future conditions (Day 1985), ESP is a scenario-based prediction technique that is often adopted in operational hydrology for streamflow forecasts. In brief, ESP generates streamflow ensemble predictions by inputting climate forcing data sampled from the observation data to a hydrologic model given the same initial condition at the time of the forecast and thus is sometimes referred to as 'conditional Monte Carlo simulation' (Day 1985). The initial conditions in the ESP framework which can have a substantial effect on the prediction outcomes, such as soil moisture, are often estimated through observations just before the time of forecast.

By far, ESP has been adopted in the field of water resources engineering widely for various purposes. A popular approach for utilizing ESP is in reducing different types of uncertainties inherent in most hydrologic modeling (e.g., input uncertainty, parameter estimation uncertainty, etc.) (Hwang et al. 2011; Tongal & Booi 2017). Furthermore, ESP is also adequate when coupled with the optimization models for deriving optimal release decisions in reservoir

operations during the formulation. In this context, multiple studies have proved that incorporating ESP may improve system performance under various spatial–temporal scales (Kim et al. 2007; Jafarzadegan et al. 2014). Other than applying ESP for streamflow prediction, the use of ESP may be extended to improving the prediction of other variables (Day 1985) such as hydrological drought, as presented in this study. Therefore, this study aims to present a methodology that utilizes the concept of ESP to conduct drought prediction, namely EDP. Further details regarding the procedures behind EDP are thoroughly explained in the following subsections.

2.2 Standardized runoff index

The proposed methodology of this study, EDP, requires the output from the ESP model (i.e., streamflow prediction ensembles) to be converted to an index that can quantify drought severity. Among multiple candidate indices, this study selected the SRI, which is one of the most commonly used indices to represent hydrological drought through the streamflow level. SRI was first developed by Shukla and Wood (2008) to consider various kinds of streamflow inflow such as surface runoff and subsurface runoff for hydrological drought analysis. SRI is appropriate to analyze both short- and long-term hydrological droughts since it is able to analyze droughts of various durations through time scale accumulation (e.g., 1, 3, and 12 months).

A simple calculation procedure of the SRI adopted in this study is as follows. The first step is calculating the cumulative streamflow over a given time scale k by accumulation. Next, the cumulative probability values of the cumulative streamflow are estimated with the appropriate probability density function the cumulated streamflow is assumed to follow (normal distribution in this study). Finally, the SRI is derived by converting the cumulative probability into the variable which follows the standard normal distribution. Let $g(\cdot)$ be the function that calculates the SRI described from the above procedure. Then the one month ahead ensemble drought prediction (EDP_{t+1}) can be converted through Eq. (1) where q_t represents the streamflow at the time interval t , the subscript t indicates the time interval month, k is the given time scale for SRI accumulation, and μ_0 and σ_0^2 are the mean and standard deviation of the EDP, respectively.

$$EDP_{t+1} = g(q_{t-k+2}, \dots, q_t, q_{t+1}) \sim N(\mu_0, \sigma_0^2) \quad (1)$$

2.3 Bayes' theorem

Bayesian inference enables the update of prior distribution with the introduction of novel information through observation with a predefined likelihood function. Mathematically,

the equation for the Bayes' theorem consists of the following three key components: prior distribution, likelihood function, and posterior distribution, as shown in Eq. (2).

$$p(D|X) = \frac{p(X|D)p(D)}{p(X)} \quad (2)$$

Where D is the random variable of interest (i.e., SRI in this study), X is the new information for the random variable of interest (i.e., soil moisture in this study), $p(D)$ is the prior distribution, $p(X|D)$ is the likelihood function, $p(X)$ is the marginal distribution of X , and $p(D|X)$ is the posterior distribution to be updated. As the conventional ESP framework is prone to how the initial conditions before the simulation are set, this study aimed to improve the EDP by reflecting the soil moisture information via the Bayes' theorem. More detailed explanations regarding each component in Eq. (2) are in the following subsections.

2.3.1 Prior distribution

In this study, the prior distribution $p(D)$ is estimated from the distribution of the EDP, calculated from the SRI values of different time scales. Since the EDP is an ensemble of drought indices that is assumed to follow normal distribution, the representative statistics of D can be expressed as Eq. (3), where μ_0 and σ_0^2 are the mean and variance of the EDP, respectively.

$$p(D) \sim N(\mu_0, \sigma_0^2) \quad (3)$$

2.3.2 Likelihood function

The likelihood function is the conditional probability of X given D , where the random variable X is the soil moisture index (SMI) which is the satellite observation data being provided by APEC Climate Center (APCC) since 2001. In other study cases, the likelihood function was often estimated from the past performance of the ensemble model (Luo et al. 2007; Seo et al. 2019). However, in this study, the likelihood function was obtained from the linear regression model between D (SRI) and X (SMI) to update the prediction results with soil moisture information, as shown in Eq. (4).

$$X_t = b_0 + b_1 D_{t+1} + \varepsilon \quad (4)$$

Where the subscript t is the time interval month, and b and ε are the regression parameters and residuals, respectively. The residual ε follows a normal distribution with zero mean and standard deviation σ_ε . As a result, the likelihood function adopted in this study can be expressed as Eq. (5).

$$p(X_t|D_{t+1}) \sim N(b_0 + b_1 D_{t+1}, \sigma_\epsilon^2) \quad (5)$$

2.3.3 Posterior distribution

Based on Bayes' theorem, the posterior distribution can be estimated using normal distribution as Eq. (6) when both the prior distribution and the likelihood function follow a normal distribution as well (Lee 1997). The parameters of the posterior distribution can also be easily derived through equations Eq. (7–8), which is a kind of variance-weighted average of the prior distribution and the likelihood function.

$$p(D_{t+1}|X_t) \sim N(\mu_p, \sigma_p^2) \quad (6)$$

$$\frac{1}{\sigma_p^2} = \frac{1}{\sigma_0^2} + \frac{b_1^2}{\sigma_\epsilon^2} \quad (7)$$

$$\frac{\mu_p}{\sigma_p^2} = \frac{\mu_0}{\sigma_0^2} + \frac{b_1^2}{\sigma_\epsilon^2} \left(\frac{X_t - b_0}{b_1} \right) \quad (8)$$

Where μ_p and σ_p^2 are the mean and standard deviation of the posterior distribution.

3 Case study

3.1 Study area and meteorological data transformation

The proposed methodology was applied to eight major basins located in South Korea (Fig. 1), all with more than 30 years of observation data sets since it is generally considered stable when the drought indices are generated with observations for more than 30 years. All eight dams are multi-purpose reservoirs that are responsible for more than one of the following objectives: hydroelectric power generation, agricultural water supply, industrial water supply, and environmental flows. As in Fig. 1 and Table 1, the eight dams are all distinctive in terms of their location, capacity, and scale. Thus, a comparative analysis of the dams regarding their characteristics can also be conducted during the evaluation process of this study.

For the eight dams, historical meteorological data sets including daily precipitation, temperature, and daily inflow were collected from either the Korea Meteorological Administration or the Korea Water Cooperation (K-water). The daily potential evapotranspiration was estimated by the FAO Penman–Monteith equation No. 56 method (Allen et al. 1998), and the meteorological data sets were converted into the areal mean values for each basin through the Thiessen polygon method.

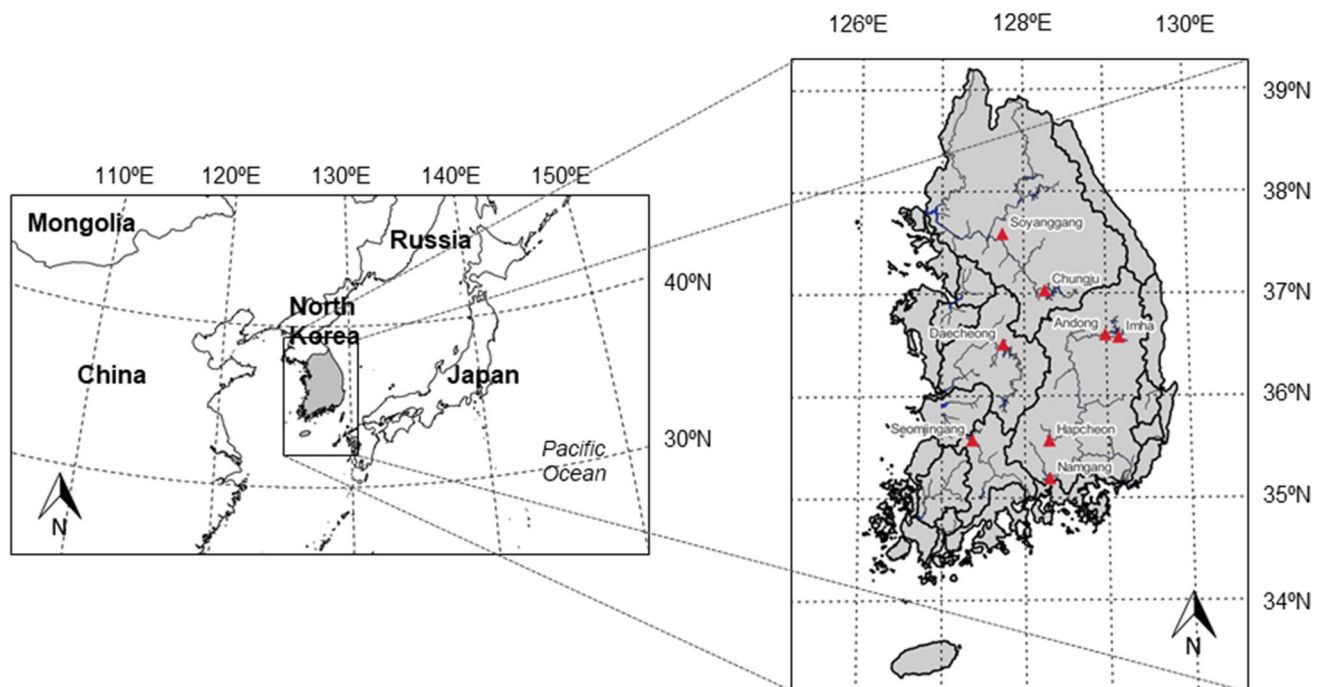


Fig. 1 Map of the study area and the location of eight dams

Table 1 Specific information about the eight dams analyzed during this study

Number	Dam Name	Drainage Area (km ²)	Annual Inflow (10 ⁶ m ³)	Observation Period
1	Soyang	2,703	2,148	1973–2017
2	Daecheong	3,204	2,722	1981–2017
3	Andong	1,584	950	1977–2017
4	Sumjin	763	502	1975–2017
5	Chungju	6,684	4,872	1986–2017
6	Hapcheon	925	573	1989–2017
7	Namgang	2,285	2,031	1976–2017
8	Imha	1,361	545	1992–2017

For the rainfall-runoff model which is an essential component in the ESP framework, this study used the TANK model, a conceptual model that describes the rainfall-runoff process in a structure that consists four tanks (Sugawara 1995). To consider the snow accumulation–melting, the modified TANK model by McCabe and Markstrom (2007) was used, and the parameters of the TANK model were calibrated and validated using the observation inflow data of each basin from 1976 to 2015, as suggested in the previous study by Seo and Kim (2018), which used the shuffled complex evolution algorithm with the observation data. For more specifics regarding the steps for parameter estimation in the TANK model, the reader is referred to Seo and Kim (2018).

Finally, to evaluate the prediction results in terms of both short- and long-term droughts, this study selected the two-time scales of 3 and 12 for the monthly SRI values, since the SRI3 and SRI12 are known as the appropriate drought indices for the short- and long-term hydrological drought in South Korea (Son et al. 2011). Finally, the drought phases were categorized by the corresponding to the SRI values as shown in Table 2 to distinguish the severity of droughts during the evaluation process.

3.2 EDP model development

Four types of EDP models (EDP, EDP + A, EDP + S, and EDP + AS) were generated using the methodological process described in the previous sections. First, the EDP model is a conventional model which only includes the information from SRI ensembles. The EDP + A model includes the information from the precipitation forecast from the APCC to update the former EDP model. During this process, the pdf-ratio method (Stedinger and Kim 2010) which revises the probability of each meteorological scenario (i.e., input ensemble) based on the probabilistic climate forecast, is used to reflect novel climate information (i.e., precipitation forecasts). To update the EDP model based on soil moisture, the

Table 2 Drought classification according to the SRI values

Values	Phase	Interval Probability	Cumulative Probability
$0 < \text{SRI}$	No Drought	0.500	1.000
$-1 < \text{SRI} \leq 0$	D0	0.341	0.500
$-1.5 < \text{SRI} \leq -1$	D1	0.092	0.159
$-2 < \text{SRI} \leq -1.5$	D2	0.044	0.067
$\text{SRI} \leq -2$	D3	0.023	0.023

lag-1 correlation coefficients between SRI and SMI were calculated and confirmed.

The next two models, EDP + S and EDP + AS models both update the prior distributions of either the EDP or EDP + A forecasts through the Bayes' theorem with the information of SMI satellite observation data from the APCC as likelihood function. As previously mentioned, the likelihood function is estimated by the time series regression between two random variables, SMI and the distribution of EDP as shown in Eq. (4). The overall flow chart of this study adopted during the model formulation process is shown in Fig. 2.

3.3 Performance evaluation metrics

Among multiple candidates of evaluation metrics, this study selected three metrics for comparative evaluation among the prediction models: root mean squared error (RMSE), ranked probability skill score (RPSS), and the Brier skill score (BSS). Whereas the RMSE measures the accuracy of the prediction models in comparison with the observed data, the other two metrics, RPSS and BSS, are both score-based metrics that are often adopted when evaluating probabilistic predictions.

The first metric, RMSE, was used to evaluate the quantitative error between the true data and the single-valued prediction. In the RMSE, both bias and variability of predicted values are included. Ideally, zero RMSE across the prediction horizon is favorable since it indicates that the model predictions are able to forecast perfectly identical to the observed data. Specifically, RMSE is computed by taking the root of the mean squared error as written in Eq. (9) where $\bar{P}_i$ is the mean of the prediction ensemble at time step i , O_i is the observed value at time step i , and N is the total number of time steps.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\bar{P}_i - O_i)^2} \quad (9)$$

The ranked probability skill score (RPSS) and the Brier skill score (BSS) were used to evaluate the predictability of the drought occurrence either in multi-categorical

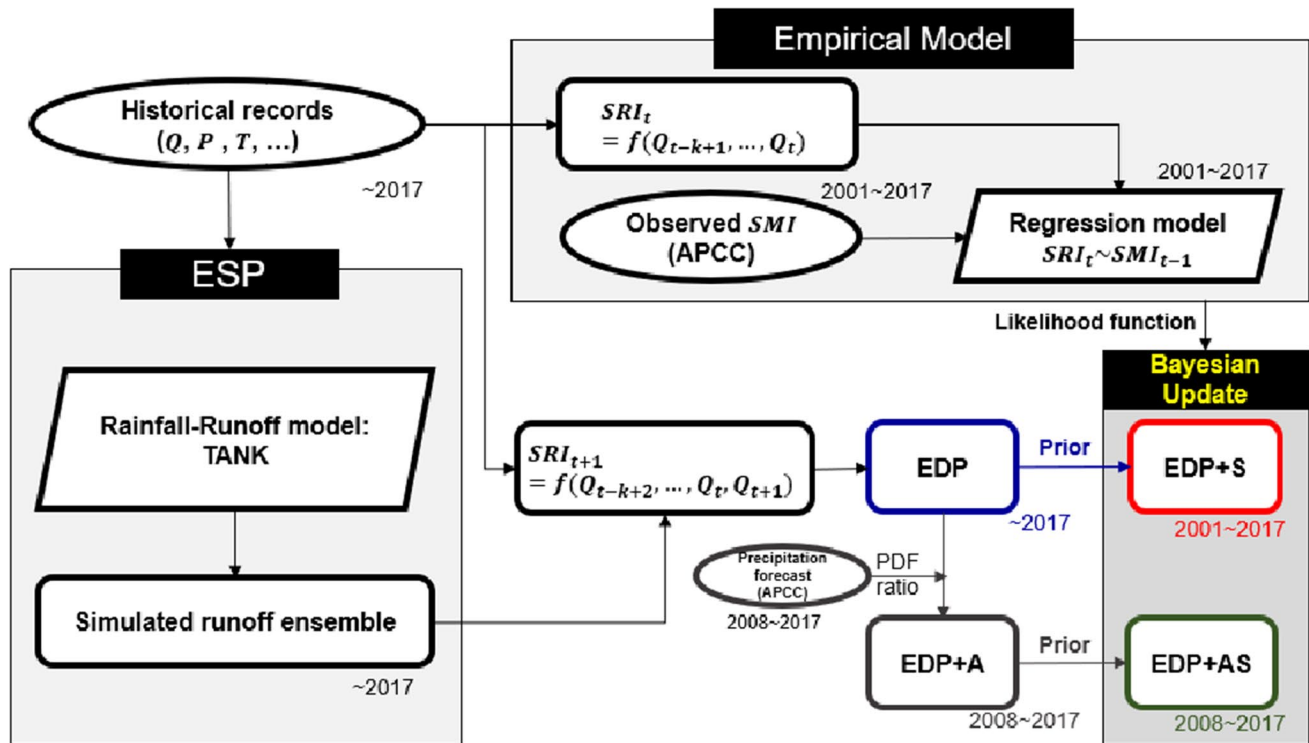


Fig. 2 Flow chart of the EDP model updated with the Bayes' Theorem

or binary outcomes. They are both skill scores which are similar to the rank probability score (RPS), which penalizes the prediction results that are far from actual with a greater penalty and is often adopted in the field of climatology for forecast evaluation. RPS is calculated from the sum of the mean squared difference between the category of the forecast and the actual category the observation belongs to. Ideally, a zero RPS value is expected for a perfect forecast, and the greater the RPS value, the poorer the forecast. RPSS can then be computed from the RPS value according to Eq. (10), where RPS_{cl} represents a reference forecast from the climatological predictions.

$$RPSS = 1 - \frac{RPS}{RPS_{cl}} \quad (10)$$

Finally, the concept of BSS is derived from the Brier score (BS), which is sometimes referred to as the reduced version of the RPSS. Both the BS and the BSS are identical to the RPS and the RPSS in cases in which the forecasts can be categorized into exactly two categories (i.e., drought or non-drought) (Weigel et al. 2007). In this specific case study, BSS is used to evaluate whether the drought prediction model can predict drought occurrences above a certain phase. Ultimately, this study utilized the three metrics to evaluate the performance of the forecast results from various EDP models comprehensively.

4 Results and discussion

4.1 Determination of the EDP models

All four EDP models (EDP, EDP + S, EDP + A, and EDP + AS) were generated to forecast the 1-month lead droughts for the eight basins from 2008 to 2017. After fitting the prior distribution with the simulated ensembles from the TANK model, the applicability of implementing SMI into the SRI was validated using the linear regression coefficient between SRI and SMI. As in Table 3, the lag-1 correlation coefficients between SRI and SMI of all eight basins surpassed the critical value of 0.136 at a significance level of 95%. This result indicates that the inclusion of soil moisture information in terms of likelihood function is valid and applicable. The parameters of the likelihood function were then estimated with the average values among the four folds after going through the fourfold cross validation method. For more specific details regarding the parameter estimation procedures of the likelihood function, refer to Kim (2020).

Other than the EDP + S model, which updates the former EDP model with SMI, probabilistic precipitation forecasts were also included to create the two other models: EDP + A and EDP + AS. Whereas the EDP + A model utilized novel climate information via the pdf-ratio method from the distribution of the EDP model forecasts, the EDP + AS model includes both the information from the soil moisture and

the precipitation forecast for improved prediction accuracy. For illustration, Fig. 3(a) shows an example of implementing four EDP models to predict short-term drought (i.e., using SRI3) at the Soyang basin in July 2014. The light blue colored bars represent the ensemble members of the SRI and the grey line indicates the likelihood function. Blue, red, black, and green line represents the predicted distributions of the EDP models: EDP, EDP + S, EDP + A, and EDP + AS, respectively. Figure 3(b) shows the time series plot during the entire prediction horizon of the four models, using the expected value of the distribution as the representative value. The framework used to determine the distributions of the four models was identically applied to the eight basins, and further analysis of the performance of each model are to be included in the following subsections.

4.2 Drought prediction performance evaluation

For further analysis, this study calculated the performance metrics in terms of length of drought (i.e., short- and long-term), and the period of prediction (i.e., irrigation and

non-irrigation) as the annual precipitation pattern may vary in the location of the target basins as shown in many previous studies regarding droughts (Kim et al. 2019, 2021, 2022). Moreover, as the droughts are known to heavily affect the irrigation industry in Korea, this study distinguished the irrigation period (May through September) and non-irrigation period (October through April of the following year) accordingly. This study predicted short-term droughts since they are considered important due to the following reasons, especially in the study area: (1) the capacity inflow ratios of the reservoirs are close to 1 in many reservoirs in Korea, which makes the long-term planning and management challenging, and (2) the precipitation patterns in Korea are concentrated in the 3-month monsoonal period, which increases the need for short-term management based on short-term predictions even more important.

4.2.1 Short-term drought prediction

In terms of short-term drought prediction, the performance of the models was calculated from the SRI3 values. As in Table 4, the RMSE values of the prediction results mostly surpassed 0.5, except for the Sumjin basin indicating that short-term drought predictions are challenging. The RMSE values of all basins tended to increase especially during the irrigation period, in which the increased variability of precipitation hinders accurate predictions compared to the non-irrigation periods. Figure 4 shows the changes in RMSE values from the EDP model for short-term drought predictions across the basins as a heatmap. In Fig. 4, red colors indicate improvements from the EDP model (favorable outcomes), and the blue colors consequently indicate unfavorable outcomes. As a comprehensive result from Fig. 4, the two models that included the soil moisture information to update the

Table 3 Lag-1 correlation coefficients between SRI and SMI

Number	Dam Name	SRI3	SRI12
1	Soyang	0.489	0.260
2	Daecheong	0.516	0.219
3	Andong	0.519	0.262
4	Sumjin	0.473	0.155
5	Chungju	0.506	0.288
6	Hapcheon	0.439	0.142
7	Namgang	0.436	0.155
8	Imha	0.474	0.270

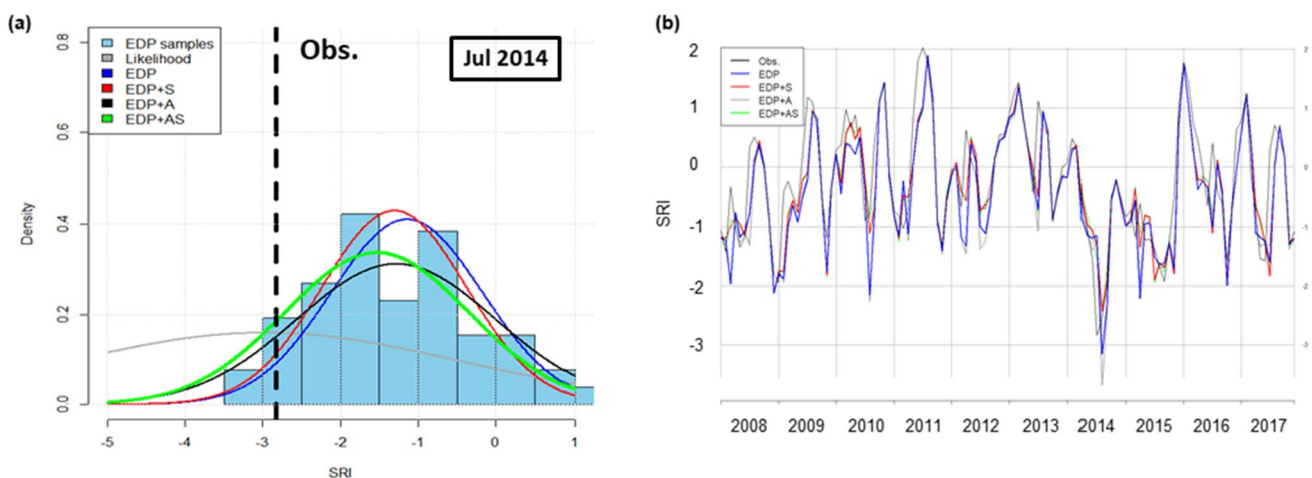


Fig. 3 Examples of the EDP model formulation for short-term drought at the Soyang basin: (a) distribution of the EDP models for July 2014; (b) monthly time series plot for the expectation values of EDP model distributions

Table 4 Average RMSE values of one-month lead forecasts for short-term droughts

Model	Soyang	Daecheong	Andong	Sumjin	Chungju	Hapcheon	Namgang	Imha
EDP	0.608	0.593	0.563	0.491	0.741	0.486	0.502	0.633
EDP+S	0.528	0.586	0.538	0.491	0.657	0.5	0.502	0.624
EDP+A	0.624	0.604	0.578	0.504	0.762	0.488	0.514	0.594
EDP+AS	0.523	0.595	0.564	0.504	0.647	0.504	0.513	0.624

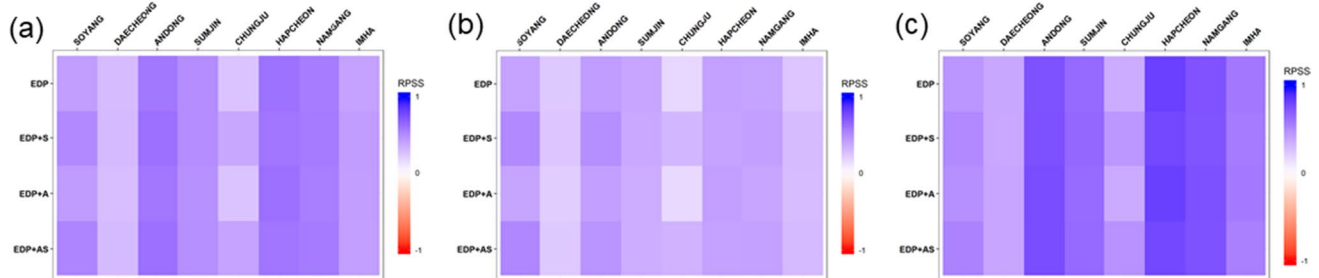
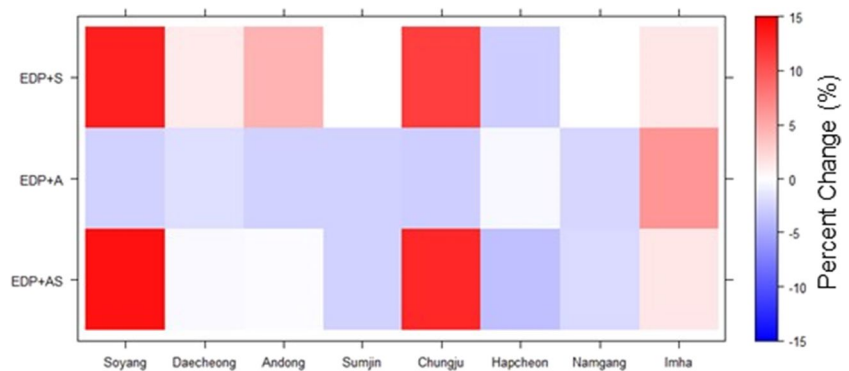
conventional EDP model (EDP+S and EDP+AS models) resulted in an improvement in prediction accuracy measured in RMSE. Especially, the Chungju and Soyang basins (ranking first and third in terms of basin drainage area) showed the most significant improvements with 11.34%, and 13.16% increases in the EDP+S model, and 13.98%, 12.69% increases in the EDP+AS model, respectively. This indicates that in terms of short-term drought predictions, the inclusion of soil moisture is more effective in basins with larger drainage areas.

In terms of probabilistic evaluations of short-term droughts, measured with RPSS and BSS values, the results are shown in Figs. 5 and 6. From Fig. 5, in which the RPSS values of the four EDP models are shown in terms of heatmap, it can be inferred that the predictions from all four models improve the forecast results using only climatological values. The improvements can be especially highlighted during the non-irrigation periods (Fig. 5c), in which the intertemporal variabilities of precipitation

are relatively smaller than the other two periods. On the other hand, when the short-term drought predictions are measured with BSS with respect to D0 and D2 drought phases as in Fig. 6, all EDP models were not able to predict extreme droughts with accuracy, especially in Daecheong and Chungju basins. On average, the EDP+S model showed the most improvement from with a BSS score of 0.366, and the EDP+AS model followed the EDP+S model with a score of 0.353. This result implies that although extreme droughts are difficult to predict, the inclusion of soil moisture information during the predictions may enhance the prediction accuracy, especially during the non-irrigation periods.

4.2.2 Long-term drought prediction

Similarly from the short-term drought predictions, the long-term droughts were evaluated with the same metrics but with the SRI12 values. Table 5 and Fig. 7 show the results of the

Fig. 4 Heatmap for the improvement of RMSE values of alternative EDP models across the basins for short-term drought predictions**Fig. 5** The RPSS heatmap of the four EDP models for one-month lead short-term drought predictions during (a) all, (b) irrigation, and (c) non-irrigation periods

long-term drought predictions measured in terms of RMSE. Comparing the results from short-term droughts, predicting long-term droughts yielded more reliable accuracy. The primary reasons behind this outcome can be derived as the prominent trade-off relationship between volatility and prediction accuracy, as SRI12 is an accumulated index of summation of streamflow for 12 months. Furthermore, the prediction performance was more sensitive with respect to basins, rather than the models and the information included

during model formulations. As in Fig. 7, the improvement of RMSE does not stand out as the short-term predictions, as the predictions from the conventional EDP model for long-term drought predictions yielded reliable outcomes in terms of RMSE.

The probabilistic evaluation results of the long-term droughts are shown in Fig. 8 (RPSS) and 9 (BSS). In accordance with the results measured in terms of RMSE, long-term drought prediction evaluations yielded more

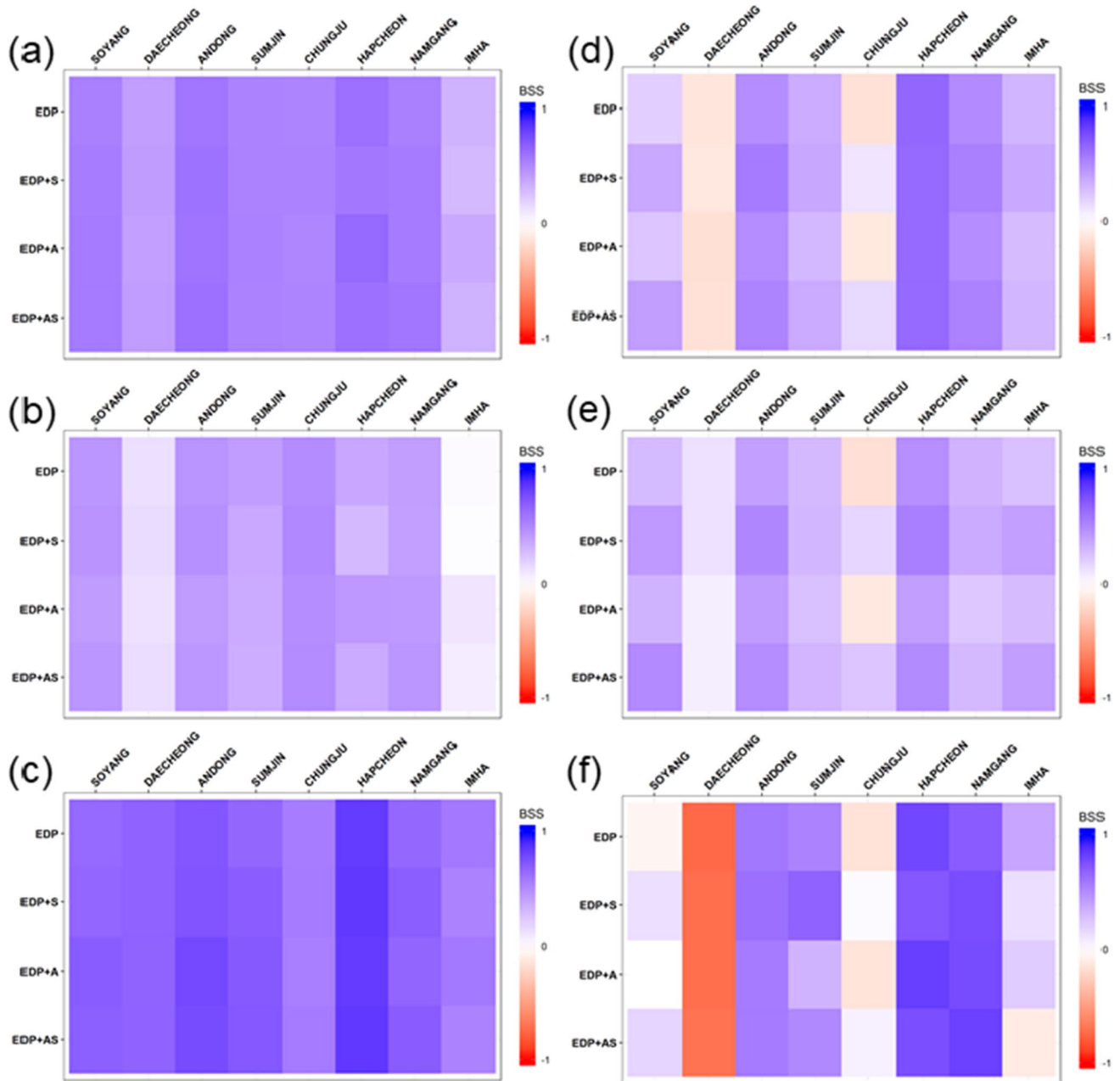


Fig. 6 The BSS heatmap of the four EDP models for one-month lead short-term drought predictions above D0 phase drought occurrence predictions during (a) all, (b) irrigation, (c) non-irrigation periods;

and above D2 phase drought occurrence predictions during (d) all, (e) irrigation, and (f) non-irrigation periods

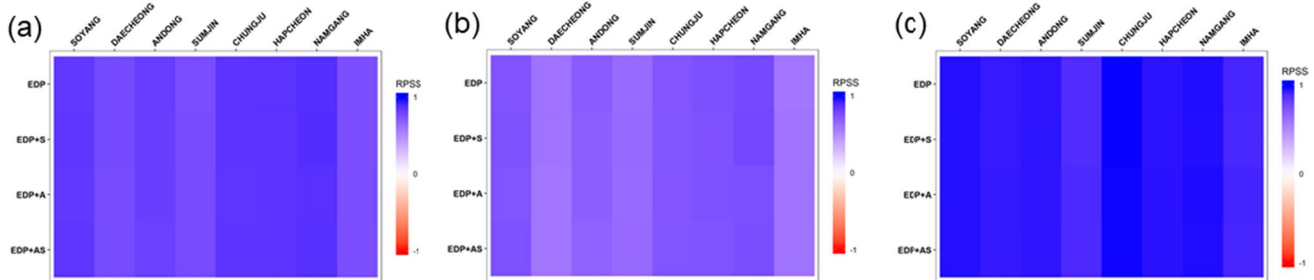
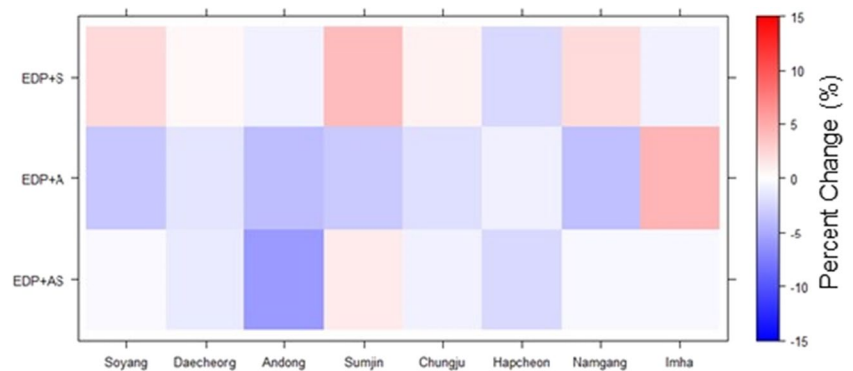
Table 5 Average RMSE values of one-month lead forecasts for long-term droughts

Model	Soyang	Daecheong	Andong	Sumjin	Chungju	Hapcheon	Namgang	Imha
EDP	0.278	0.258	0.236	0.255	0.264	0.222	0.242	0.253
EDP+S	0.272	0.257	0.238	0.245	0.262	0.227	0.237	0.255
EDP+A	0.287	0.262	0.245	0.263	0.269	0.224	0.251	0.242
EDP+AS	0.279	0.261	0.25	0.252	0.266	0.227	0.243	0.254

reliable outcomes as well. As notable in Fig. 8c, Fig. 9c and f, the prediction of long-term droughts during the non-irrigation periods with smaller intertemporal variabilities resulted in the best prediction results in most of the basins. Among different EDP models, notable differences or improvements in terms of either RPSS or BSS were not observed. Thus, it can be concluded that the inclusion of additional information (i.e., either precipitation forecasts in the EDP+A models or soil moisture information in the EDP+S models) affects the prediction accuracy of short-term droughts more significantly. This implies that the applicability of the proposed methodology of various EDP frameworks is more suitable for short-term drought predictions.

5 Conclusion & discussions

This study developed various EDP models in order to perform probabilistic prediction of hydrological droughts using the key components of the conventional ESP framework. During the model formulation process, precipitation forecasts and soil moisture were included in the basic EDP model to analyze the effect of including additional hydrological variables in prediction accuracy. The proposed methodology was applied to eight basins located in Korea, all with more than 30 years of accumulated observation data. For evaluation, both deterministic and probabilistic metrics were utilized and the prediction results were analyzed under the cross-validation process with the observation data. The key findings from this study can be summarized as follows:

Fig. 7 Heatmap for the improvement of RMSE values of alternative EDP models across the basins for long-term drought predictions**Fig. 8** The RPSS heatmap of the four EDP models for one-month lead long-term drought predictions during (a) all, (b) irrigation, and (c) non-irrigation periods

1. The alternative models proposed in this study (EDP + A, EDP + S, EDP + AS) were proven to improve prediction accuracy, especially in terms of short-term drought predictions.
2. The prediction of droughts with soil moisture (EDP + S and EDP + AS) notably improved in basins with larger drainage areas, which can lead to the necessity of the inclusion of soil moisture in basins with larger drainage areas.

3. When predicting long-term droughts, the results yielded that predicting droughts with smaller intertemporal variabilities (i.e., non-irrigation periods) proved to be more effective.

Despite multiple notable findings and the improvements of the probabilistic drought prediction methodology proved in this study, few potential advances regarding both the methodological components and the data remain. First of

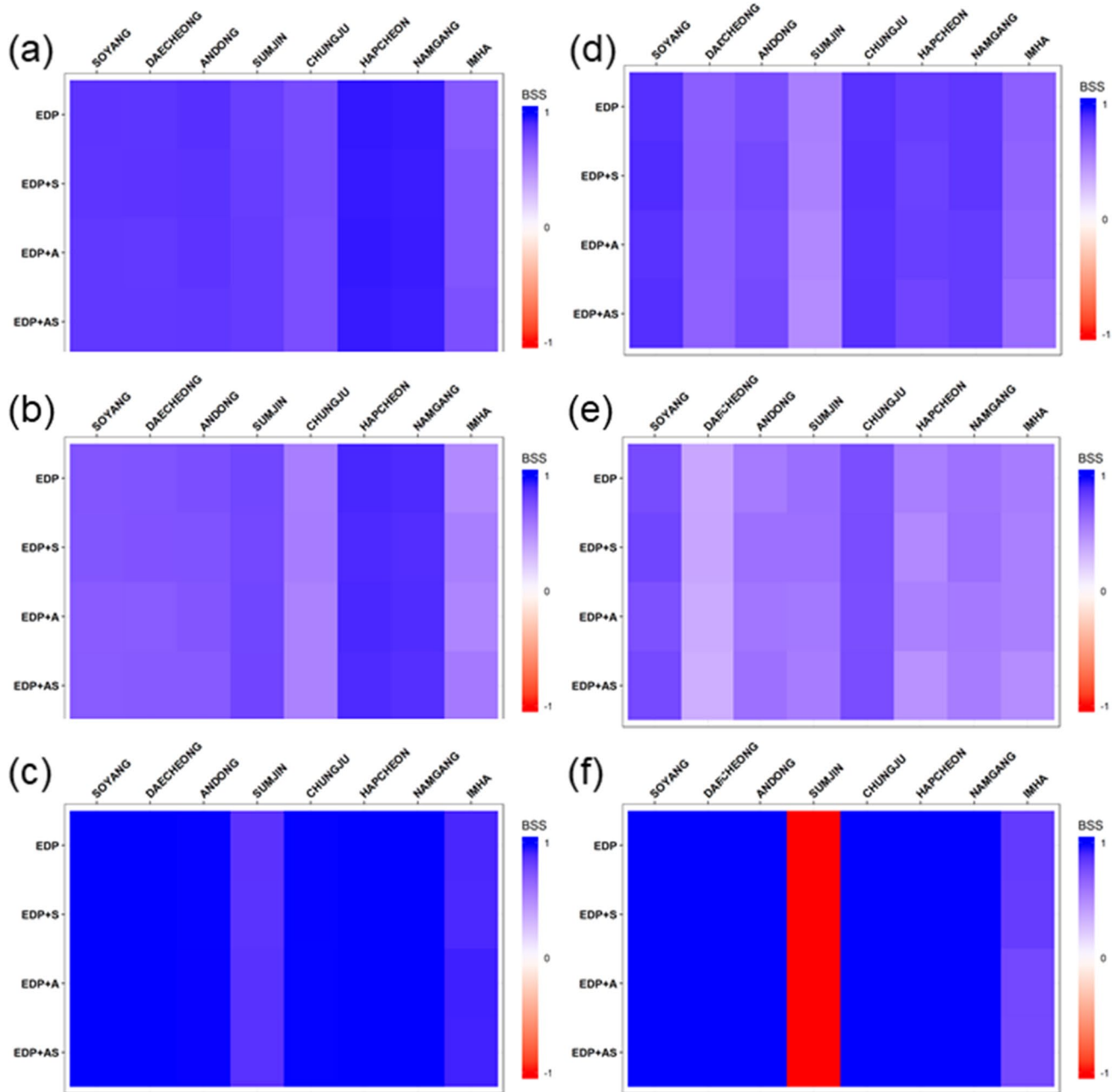


Fig. 9 The BSS heatmap of the four EDP models for one-month lead long-term drought predictions above D0 phase drought occurrence predictions during (a) all, (b) irrigation, (c) non-irrigation periods;

and above D2 phase drought occurrence predictions during (d) all, (e) irrigation, and (f) non-irrigation periods

all, expanding the spatial scope from local to a global scale is possible by using various ensemble forecasts from other agencies. For instance, using the datasets from the European Centre for Medium-Range Weather Forecasts and National Center for Atmospheric Research provide global forecasts for the streamflow ensemble and soil moisture will prove the applicability of the proposed framework to a global scale when appropriate downscaling and bias-correction procedures were preceded.

Another direction includes the use of robust machine learning techniques (i.e., long short-term memory models for streamflow prediction and Gradient Boosted Decision Trees for drought classifications) during the application process. Since many studies suggest the use of machine learning techniques for hydrologic predictions accompanied with physics-based hydrological models (Gurbuz et al. 2023), the framework proposed in this study holds promise when coupled with machine learning approaches.

Authors' contributions Dae Ho Kim, and Young-Oh Kim contributed to the study conception and design. Material preparation, data collection and analysis were performed by Dae Ho Kim and Gi Joo Kim. The first draft of the manuscript was written by Gi Joo Kim and Dae Ho Kim. Gi Joo Kim and Young-Oh Kim commented on previous versions of the manuscript. Gi Joo Kim, Dae Ho Kim and Young-Oh Kim read and approved the final manuscript.

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Declarations

Competing interests The authors have no conflicts of interest to declare that are relevant to the content of this article.

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